Structured Anchors Predict, Random Probes Calibrate: A Per-Image Conformal Layer for Sparse-Guided Depth

Anonymous CVPR submission

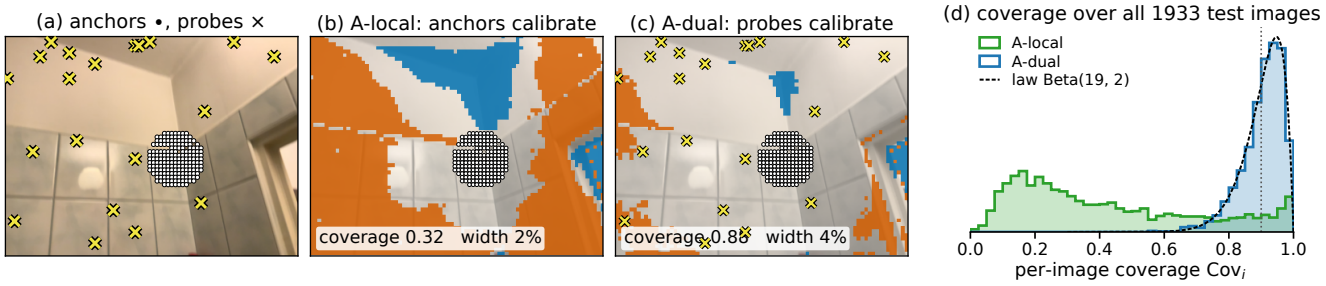


Figure 1. **Reusing structured depth measurements for calibration fails silently; separating their roles does not.** A frozen monocular model (UniDepthV2) plus 200 metric measurements on an ARKitScenes frame: 180 placed by the sensor in one cluster (•) and 20 uniformly random probes (×). (b) Calibrating within the image on the clustered measurements gives intervals 2% wide that hold near the cluster and miss the true depth over most of the frame (vermillion: truth farther than the interval, blue: nearer; dark grey: no ground truth). (c) A-dual lets the cluster correct the prediction and only the probes calibrate it. (d) Over all test images, A-local's per-image coverage scatters far below the 0.9 target (dotted), while A-dual's follows the law of Proposition 1. The frame has A-local coverage near its median over test images.

Abstract

Monocular metric depth foundation models predict metres at every pixel but say little reliable about their error, and much of that error is one scale factor per image. A few sparse metric measurements, as delivered by a phone LiDAR or a few-beam scanner, remove most of it; we call this setting sparse-guided monocular depth. Conformal calibration inside each image can then attach intervals with a per-image coverage guarantee, but only if the measurements are placed uniformly at random. Real sensors place them on scan lines, in clusters, on edges or only at short range, and such structured measurements, reused for calibration, silently void the guarantee: with clustered measurements, per-image calibration covers only 0.31–0.47 of pixels at a 0.9 target. We separate the two roles. Structured anchors that follow the sensor build the prediction; a small budget of uniformly random probes, never seen by the predictor, sets a per-image conformal threshold. The per-image guarantee of this calibration layer holds for any anchor layout and any probe-independent centre without calibration images, and the number of probes sets its spread across images. On NYU and KITTI with three frozen models and five layouts it reaches nominal coverage in every setting and is 16–69% narrower than a hull-based per-image baseline, with a lower interval score in 46 of 60 settings, the exceptions mostly under clustered anchors. On ARKitScenes the layer runs unchanged on a depth-completion network, the better centre under uniform anchors, while the frozen model with our anchor-built centre is better under clustered anchors. With iPad LiDAR checked against laser scans, probes certify what they measure: laser probes certify the depth even when the anchors are iPad readings, whereas iPad probes certify the iPad reading, and under uniform anchors their coverage of the depth falls well below the target.

1. Introduction

An AR device placing content on a table, or an indoor robot planning around furniture, can run a monocular metric depth foundation model [9, 23, 34, 52] next to a cheap sparse depth sensor: a phone LiDAR, a time-of-flight module, a few-beam scanner or SLAM points. Such a system needs, for each frame, an interval on every pixel's depth that holds for that frame, not on average over a dataset.

Conformal calibration inside the frame, on the sparse measurements, promises exactly this [21, 30], but only if the measurements are placed uniformly at random. Real sensors return scan lines, clusters, depth edges, or only nearby surfaces. When such structured measurements both correct the prediction and calibrate it (Fig. 1), nearby measurements agree, the calibration sees small errors, and the intervals become *narrower* while missing most of the frame: with clustered measurements, coverage drops to 0.31–0.47 at a 0.9 target. Only dense ground truth reveals this.

The sparse sensor is worth keeping. Native confidences are not reliable [12], and calibration across images gives only an average guarantee that hides images with far larger errors: an image's global scale error explains 63–79% of the image-to-image variation in interval coverage on NYU and KITTI. Aligning to a few metric points [38, 51] removes most of it: without anchors, split conformal prediction on a frozen model has several times the interval score of the anchored methods (Sec. 6.4). We call this setting *sparse-guided monocular depth*: a frozen foundation model supplies a dense prior, sparse metric measurements correct it, and calibration states what uncertainty remains. It overlaps with depth completion [28, 53]; our concern is the calibration step, and a completion network is one possible centre.

Idea. We give the sparse-depth budget two roles (Fig. 2). *Prediction anchors* follow the sensor layout, whatever it is, and build a local centre and scale around the frozen prediction. *Calibration probes*, a few measurements placed uniformly at random and never shown to the predictor, set one conformal threshold per image. Because centre and scale are fixed before the probes are revealed, a rank argument gives a per-image coverage guarantee, in expectation over the probe draw, for any anchor layout and any centre that ignores the probes, including a depth-completion network, without calibration images. Its spread across images follows an exact finite-population beta-binomial law, approximated by a Beta law, so the number of probes is a design choice that trades sensor budget for per-image reliability. The guarantee certifies whatever the probes measure: for sensor readings, the readings.

The statistical ingredients are known: fitting and calibration sets inside one instance [21], within-field exchangeability [30], design-based conformal prediction [50], and the conditional coverage law of split conformal prediction [47]. Our contribution is their deployment-aware composition for sparse-guided dense depth, an analysis of the sensor budget, and a validation with an independent sensor:

- **Failure.** Structured sparse depth improves the prediction but, reused for calibration, voids the per-image guarantee; coverage falls short under scan-line, clustered and near-only layouts, with misses far from anchors (Sec. 6.1).
- **A calibration layer.** Separating prediction anchors from calibration probes (A-dual, Sec. 4) gives a per-image guarantee for any layout and any probe-independent centre; the probe budget sets the per-image spread, and a hull with distance-binned calibration adds a guarantee by distance to the nearest anchor (A-dual-both). We evaluate three frozen models and five layouts (Secs. 6.2 and 6.3).
- **Which centre, and what is certified.** On ARKitScenes [7] a depth-completion network [53] is the better centre under uniform anchors, the frozen model with our anchor-built centre under clustered ones (Sec. 6.4). With iPad LiDAR checked against laser scans, the layer certifies what the probes measure: laser probes certify the depth even with iPad anchors, while with iPad values for both, truth coverage falls far below the target under uniform anchors (Sec. 6.5).

2. Related work

Monocular metric depth foundation models. Single-image depth [13] was long relative: MiDaS [38] predicts depth up to an unknown scale and shift, recovered per image by least squares. Large training mixtures and camera-aware designs have since made it metric and zero-shot: Depth Anything V2 [52] with metric finetunes, UniDepth(V2) [33, 34], Metric3D v2 [23] and Depth Pro [9]. We use them as frozen black boxes whose remaining error has a large per-image scale component.

Depth completion and sparse-guided depth. Depth completion predicts dense depth from RGB and sparse depth, from early learned models [29, 45] and spatial propagation [11, 32] to zero-shot methods: OMNI-DC [53] trains on synthetic data with diverse sparse patterns, Prompt Depth Anything [28] conditions a foundation model on a low-resolution LiDAR map, Marigold-DC [46] guides a diffusion depth prior with sparse points, and Prior Depth Anything [49] fuses metric priors of any sparsity pattern with a relative-depth model. Uncertainty-aware completion [15, 37] learns a per-pixel uncertainty, with no coverage guarantee. Closer to our centre, VI-Depth [51] aligns a monocular prediction to sparse visual-inertial depth with a global scale and shift, then a learned dense scale map, and MRAC [40] gates sparse anchors with a Theil–Sen fit and a median-absolute-deviation test against multipath outliers before a residual head on a frozen relative-depth model; both output point estimates. A-dual does not compete with these methods: it wraps whichever centre is best, and we run it on OMNI-DC as well as on frozen models.

Conformal prediction for images. Split conformal prediction [1, 27, 48] and conformal risk control [3] give finite-sample guarantees from exchangeable calibration data. For dense outputs, image-to-image regression [2], K-RCPS [43]

and conformal segmentation [31] calibrate across images, so their guarantees average over images. Kandinsky conformal prediction [4, 10] pools scores of similar regions across calibration images, a guarantee between image-wise and pixel-wise; ours is per image and needs no calibration images. Distribution-free conditional coverage is impossible in general [5, 47]; Mondrian calibration [48], localized conformal prediction [20] and covariate-class guarantees [19] recover it for chosen groups, as our distance-binned component does. Without exchangeability, coverage degrades with the distance from it [6]; structured anchors are such a departure, and we restore exchangeability by design rather than bound the loss or reweight [44]. Vovk [47] also gives the law of split-conformal coverage conditional on the calibration set, of which our probe-budget analysis is the finite-population form.

Conformal prediction inside one instance. Our per-image guarantee calibrates on measurements from the test image itself. Conformalized matrix completion [21] splits the observed entries of one matrix into a fitting set and a calibration set, with a weighted variant for non-uniform sampling; this is the statistical template of our dual roles. Within a single spatial field, randomly sampled locations are exchangeable with unsampled ones [30], and local spatial conformal methods [24] build intervals from nearby observations; design-based conformal prediction [50] ties validity to the sampling design. They assume a known sampling design; we face a sensor that imposes a structured one and spend a small random budget to restore a known design. With noisy calibration labels, conformal sets cover the noisy label, which is conservative for the clean one only under dispersive noise [14]; sensor-valued probes are noisy labels with a systematic bias.

Uncertainty for depth. Bayesian, ensemble, self-supervised and gradient-based uncertainties [22, 25, 26, 36] rank pixels by difficulty without a coverage guarantee. Native confidences of feed-forward 3D reconstruction models are overconfident, and after a per-model repair about two thirds of held-out scenes still miss nominal coverage by more than five points [12]. Online risk control [16] bounds the long-run average miscoverage over a stream, with depth experiments on a model whose scale and shift are fitted per frame to a few hundred depth pixels. Any such score, or a quantile regressor [39], could serve as our scale. Sample placement has been optimized for completion accuracy, end-to-end [8] or from uncertainty [42]; we propose none.

3. Problem setup

Prediction and residuals. An image x has a finite set $\mathcal{P}$ of pixels with valid ground-truth depth $y(p) > 0$, $p \in \mathcal{P}$. A frozen monocular metric model f predicts $\hat{y}(p) = f(x)(p)$. We work with the log-depth residual

$$d(p) = \log y(p) - \log \hat{y}(p), \tag{1}$$

so a symmetric interval in d is multiplicative in depth.

Sparse metric measurements. A sensor returns k measurements per image. At a measured pixel p it reports a depth $\tilde{y}(p)$ and hence a residual $\tilde{d}(p) = \log \tilde{y}(p) - \log \hat{y}(p)$. For a perfect sensor $\tilde{d} = d$; for a real one $\tilde{d}(p) = d(p) + e(p)$ with a sensor error e. The positions follow a *layout*, a random rule that may depend on x and $\hat{y}$ but never on y. We use five: *uniform* (a uniform sample of $\mathcal{P}$), *rows* (pixels on four random image rows, a few-beam scanner), *cluster* (the k pixels nearest one random centre, a single observed patch), *edges* (uniform among edge pixels of $\hat{y}$, feature-point-like) and *near* (uniform among the 30% of pixels with the smallest $\hat{y}$, a range-limited sensor).

Budget split. Of the k measurements, $k - n$ follow the layout and form the *structured anchor* set $\mathcal{S} \subset \mathcal{P}$. The remaining n form the *probe* set $\mathcal{Q}$, drawn uniformly without replacement from $\mathcal{U} = \mathcal{P} \setminus \mathcal{S}$. Intervals are evaluated on $\mathcal{E} = \mathcal{U} \setminus \mathcal{Q}$, the pixels that are neither anchors nor probes. For a pixel p, $\delta(p)$ is its distance to the nearest measurement, divided by the image diagonal.

Where random probes come from. Probes need a sensor that can be read at random pixels: a dense or semi-dense sensor subsampled in software (phone or tablet LiDAR, time-of-flight or structured-light cameras, as in depth upsampling [7]), or a steerable scanner such as MEMS-mirror [35] or adaptive LiDAR [8]. A fixed-pattern scanner allows probes only among its returns, and the guarantee then covers only those pixels: on KITTI, those with a LiDAR return, not the sky.

Intervals and coverage. A method outputs $C(p) = [\ell(p), u(p)]$ for every $p \in \mathcal{E}$. The coverage of image i is the fraction of its evaluation pixels that are covered,

$$\mathrm{Cov}_i = \frac{1}{|\mathcal{E}_i|} \sum_{p \in \mathcal{E}_i} \mathbf{1}\{y(p) \in C(p)\}. \tag{2}$$

At level $1 - \alpha$ we distinguish three guarantees (Sec. 4 states which method gives which):

- *Marginal:* $\mathbb{E}[\mathrm{Cov}_i] \geq 1 - \alpha$ over images, measurements and calibration data; the exchangeable unit is the image, or the scene (room, drive, visit).
- *Per-image:* $\mathbb{E}[\mathrm{Cov}_i \mid x_i, \mathcal{S}_i] \geq 1 - \alpha$ for *every* image, only the probes random: no image is always undercovered.

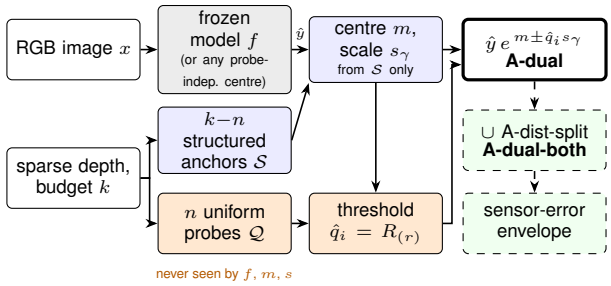


Figure 2. **Dual-role calibration.** Structured anchors $\mathcal{S}$ build a centre m and scale s_γ around a frozen prediction $\hat{y}$ (or any probe-independent centre); the n uniform probes $\mathcal{Q}$ only score it and set one threshold per image (Theorem 1). Dashed: optional stages (Theorem 2; Supp. D).

- *Per distance bin:* for bins $\mathcal{B}_b$ of δ (edges $0.02, 0.05, 0.1, 0.2$), the coverage of $\mathcal{E}_i \cap \{\delta \in \mathcal{B}_b\}$, averaged over a new scene, is at least $1-\alpha$ in expectation for every b, so far pixels are not traded for near ones.

What is covered. Every guarantee concerns the values used to calibrate: sensor readings $\tilde{y}$ certify $\tilde{y}$, not y. We always *evaluate* against the independent ground truth y.

4. Method

4.1. Architecture

In the pipeline of Fig. 2, (1) a frozen monocular model produces $\hat{y}$; nothing is retrained. (2) The sensor budget is split into $k-n$ anchors that follow the deployed layout and n uniformly random probes. (3) The anchors alone build a local centre $m(p)$ and scale $s_\gamma(p)$ around $\hat{y}$ (Sec. 4.2). (4) The probes, unseen by stages 1–3, set one conformal threshold per image (A-dual, Sec. 4.3). Optional stages add a per-distance-bin guarantee (A-dual-both, Sec. 4.5) and convert coverage of sensor readings into coverage of depth (Supp. D). Stage 1 can be any predictor that does not see the probes, *e.g.* a depth-completion network fed RGB and $\mathcal{S}$; stage 3 is then refit around its output, so A-dual is a calibration layer, not a depth estimator.

4.2. Centre and scale from structured anchors

We use the distance-aware anchor model of the baseline A-dist-split, fitted on the structured anchors only (details in Supp. E). For pixel p, the local shift $m(p)$ is one of four estimates from the anchor residuals $\tilde{d}|_{\mathcal{S}}$: the image median, the median of the 16 nearest anchors, or a Gaussian or bilateral (also weighting similarity in $\log \hat{y}$) kernel mean. The local spread $\sigma(p)$ is the median absolute deviation of the 16 nearest anchor residuals, $\delta_{\mathcal{S}}(p)$ is the distance to the nearest structured anchor as a fraction of the image diagonal, and

$$s(p) = \big(\sigma(p) + \tau\big)\, g\big(\delta_{\mathcal{S}}(p)\big), \tag{3}$$

with a floor τ and a non-decreasing g fitted by isotonic regression of normalized error against distance. The variant, bandwidths, τ, g and a cap on s are chosen per layout on held-out *head-train* images, never on calibration or test data. The only property the guarantee below needs is that m and s are functions of $(x, \hat{y}, \mathcal{S}, \tilde{d}|_{\mathcal{S}})$, *i.e.* that they do not depend on the probes.

4.3. Per-image threshold from random probes (A-dual)

When the anchors set the shift wrong everywhere, as a single cluster can, the residual is nearly flat, the threshold is set by the probes with the smallest s, and widths elsewhere grow by up to $\max s / \min s$. We therefore shrink the scale toward its per-image median $M_i = \mathrm{med}_{p \in \mathcal{P}_i}\, s(p)$:

$$s_\gamma(p) = M_i \left(s(p)/M_i\right)^\gamma, \qquad \gamma = 0.5. \tag{4}$$

s_γ depends on s only, so it is as probe-independent as s. For probes $\mathcal{Q} = \{p_1, \ldots, p_n\}$ with reported residuals $\tilde{d}(p_j)$, define the scores and threshold

$$R_j = \frac{|\tilde{d}(p_j) - m(p_j)|}{s_\gamma(p_j)}, \quad \hat{q}_i = R_{(r)}, \quad r = \lceil (n+1)(1-\alpha) \rceil, \tag{5}$$

where $R_{(r)}$ is the r-th smallest score ($\hat{q}_i = \infty$ if $r > n$). The interval at every evaluation pixel is

$$C(p) = \Big[\hat{y}(p)\, e^{\,m(p) - \hat{q}_i s_\gamma(p)},\ \hat{y}(p)\, e^{\,m(p) + \hat{q}_i s_\gamma(p)}\Big]. \tag{6}$$

Theorem 1 (Dual-role validity)**.** *Fix an image with valid pixels $\mathcal{P}$ and a structured anchor set $\mathcal{S} \subset \mathcal{P}$ chosen by any rule, which may depend on x, $\hat{y}$ and the anchor values. Let m, $s > 0$ and γ be determined by $(x, \hat{y}, \mathcal{S}, \tilde{d}|_{\mathcal{S}})$. Let $v(p)$ be the log residual of the quantity the probes report ($v = d$ for exact depth, $v = \tilde{d}$ for a sensor), defined on $\mathcal{U} = \mathcal{P} \setminus \mathcal{S}$. Draw $\mathcal{Q}$ uniformly without replacement from $\mathcal{U}$, compute $\hat{q}$ by Eq. (5) with v in place of $\tilde{d}$, and let J be uniform on $\mathcal{U} \setminus \mathcal{Q}$. Then*

$$\mathbb{P}\big(\hat{y}(J) e^{v(J)} \in C(J) \,\big|\, x, \mathcal{S}, \tilde{d}|_{\mathcal{S}}\big) \;\geq\; \frac{r}{n+1} \;\geq\; 1-\alpha, \tag{7}$$

with equality on the left when the scores have no ties and $r \leq n$. Equivalently, $\mathbb{E}[\mathrm{Cov}_i \mid x, \mathcal{S}] \geq r/(n+1)$, the expectation taken over the probe draw.

Proof sketch. Given $(x, \mathcal{S}, \tilde{d}|_{\mathcal{S}})$, every $p \in \mathcal{U}$ carries a fixed score $R(p) = |v(p) - m(p)|/s_\gamma(p)$, and the probes with J form a uniformly random $(n+1)$-subset of $\mathcal{U}$, so $R(J)$ is among the r smallest of the $n+1$ scores, hence covered, with probability at least $r/(n+1)$. Supp. A. □

This is the split argument of [21, 30] applied inside one image around a frozen prior; we claim no new theory. It needs no calibration images and uses all n probes for the

n	r	$\frac{r}{n+1}$	p10 of Cov_i	$\mathbb{P}(\mathrm{Cov}_i < 0.85)$, %
10	10	0.909	0.795	19.7
20	19	0.905	0.819	17.6
40	37	0.902	0.841	13.0

Table 1. **Probe budget at $\alpha = 0.1$, one probe draw** (Beta approximation of Proposition 1; from the definition alone, no data).

threshold ($n \geq 9$ suffices at $\alpha = 0.1$), whereas A-local-split spends half on a shift. *Scope:* the guarantee holds in expectation over the probe draw, not per region; probes must be uniform over the pixels it concerns (*e.g.* where the sensor returns); every choice, γ included, is fixed without probe values; and it covers v, what the probes measure.

4.4. Probe budget

Theorem 1 fixes the mean of Cov_i; n fixes its spread.

Proposition 1 (Per-image coverage law)**.** *Under Theorem 1 with no ties and $r \leq n$, let $N = |\mathcal{U}|$. The number of covered evaluation pixels, $(N-n)\mathrm{Cov}_i$, follows exactly the beta-binomial law* $\mathrm{BetaBin}(N-n,\, r,\, n+1-r)$. *As $N \to \infty$, $\mathrm{Cov}_i \to \mathrm{Beta}(r,\, n+1-r)$ in distribution.*

This is the finite-population form of the Beta law of split-conformal coverage conditional on the calibration set [47]; with thousands of evaluation pixels per image the Beta approximation is excellent. That measurements match this law (Sec. 6.3) is an implementation check implied by the randomization, not a finding.

Design rule. The share of images below coverage c in one draw is $\mathbb{P}(\mathrm{Beta}(r, n+1-r) < c) = \mathbb{P}(\mathrm{Bin}(n, c) \geq r)$, known before any data are seen (Tab. 1). At $\alpha = 0.1$ it falls slowly with n ($c = 0.85$: 5.5% at $n = 100$); the nominal level is the stronger lever: $n = r = 20$ (nominal 0.952) leaves 3.9% below 0.85 (measured on NYU and KITTI: 3.7–4.0%, coverage 0.952–0.953), at 31–84% more width and 8–21% more interval score than $r = 19$.

4.5. Adding a per-distance-bin guarantee (A-dual-both)

A-dual's coverage, in practice, decreases with distance to the nearest anchor. The baseline A-dist-split targets this: it fits the model of Sec. 4.2 on all k measurements and calibrates one threshold per bin of δ by conformal risk control [3] over calibration *scenes*, the loss being the miscovered fraction of bin pixels (a Mondrian construction [48]; sparse bins are merged, Supp. C). For exchangeable scenes and the same anchor rule at calibration and test, the expected bin miscoverage of a new scene is at most α in every (merged) bin. Unlike A-dual, this needs calibration scenes with dense depth from the same source as the measurements, and it certifies that source. A-dual-both is the pointwise hull $C^{\mathrm{both}}(p) = \big[\min(\ell^{\mathrm{dual}}, \ell^{\mathrm{dist}}),\ \max(u^{\mathrm{dual}}, u^{\mathrm{dist}})\big]$.

Theorem 2 (Hull validity)**.** *A-dual-both has the per-image guarantee of Theorem 1 and the per-bin guarantee of A-dist-split.*

Proof sketch. The hull contains each interval pointwise and both statements are monotone in the interval [17]. Supp. C. □

A-dual-both replaces A-dist-both, the hull of A-dist-split with A-local-split. Any hull costs width, most when the two centres disagree, as under clustered anchors.

Sensor-valued probes. With sensor-valued probes, Theorem 1 certifies the reading; an optional sensor-error envelope that converts this into an average-over-scenes guarantee on depth is given in Supp. D; it restored truth coverage here (0.993–0.996 with iPad probes) but at a higher interval score than A-dist-both.

5. Experimental setup

Datasets. *NYU Depth v2* [41]: 1,449 images from 464 rooms (Kinect, <10 m). *KITTI* [18]: 652 Eigen test plus 800 Eigen-train frames from 61 drives (Garg crop, <80 m); its ground truth comes from the LiDAR that would provide the anchors, so it cannot test an independent sensor. *ARKitScenes* [7], depth-upsampling validation split: 5,594 frames from 107 visits, ground truth from a Faro laser scan, deployed sensor the iPad Pro LiDAR depth from ARKit (fused, with confidence $\{0, 1, 2\}$), which reads about 1.6% nearer than the laser (preprocessing in Supp. G). Valid pixels are kept on a lattice (stride 4 on NYU, 2 on KITTI, 8 on ARKitScenes) and stored per image, so per-image statistics are exact. Every split is by scene (room, drive, visit): 30% head-train (fits m, s and hyper-parameters), 35% calibration (cross-scene thresholds; unused by A-dual), 35% test.

Frozen models. UniDepthV2-Large [34], Depth Anything V2 Large metric [52] (indoor checkpoint, fine-tuned on Hypersim, on NYU and ARKitScenes; outdoor checkpoint, fine-tuned on Virtual KITTI 2, on KITTI) and, on NYU and KITTI, Depth Pro [9]; Metric3D v2 [23] is in the supplement. NYU and KITTI are not in the training mixtures of these models (Supp. G). UniDepthV2's training mixture includes ARKitScenes (split not reported) [34]; our validation-split frames are likely unseen, the domain is not, which affects width, not any guarantee. Depth Pro, trained on ARKitScenes, is not used there.

Measurements. $k \in \{50, 200\}$ measurements per image, of which $n = 20$ are probes (ablation $n \in \{10, 40\}$), under the five layouts of Sec. 3. On NYU and KITTI they report the ground truth. On ARKitScenes, anchors and probes

each report the laser or the iPad value, a 2×2 factorial (laser/laser, iPad/laser, laser/iPad, iPad/iPad); coverage is always measured against the laser. For UniDepthV2 and DA-V2 (E18): $k = 200$ (180 anchors, 20 probes), uniform and cluster layouts, 20 visit splits $\times$ 10 probe draws.

Methods. All methods share positions and the log-residual score. *Baselines:* A-local [30] calibrates within the image on the structured anchors themselves (half fix a shift, half set the width; valid only for uniform anchors); A-local-split does the same on the n probes; A-dist-split (Sec. 4.5); A-dist-both = hull(A-dist-split, A-local-split). *Ours:* A-dual and A-dual-both. *Completion centre (E19):* OMNI-DC [53] (v1.1, Depth Anything V2 prior; trained only on five synthetic datasets, Hypersim, IRS, TartanAir, BlendedMVS and Virtual KITTI, so zero-shot on ARKitScenes) on ARKitScenes, $k = 200$ (180 anchors, 20 probes), uniform and cluster layouts, laser anchors with laser or iPad probes, UniDepthV2 as the frozen alternative, 3 visit splits $\times$ 1 probe draw. OMNI-DC sees RGB and the structured anchors only, never the probes. Both centres are calibrated four ways: split-CP (one quantile of $|\log y - \log \hat{y}|$ over calibration-scene pixels; marginal; needs dense laser depth), A-dist-split, A-dual and A-dual-both, with m, s refit per centre.

Metrics. Coverage is image-averaged Eq. (2); per-image spread is the 10th percentile (p10) of Cov_i and the share of *short* images, whose coverage on one probe draw is below $1 - \alpha - 0.05 = 0.85$ (law: 17.6% at $n = 20$). Width is the median of $(u - \ell)/\hat{y}$, or of $(u - \ell)/y$ when centres differ. The pixel interval score $(u - \ell) + \frac{2}{\alpha}\left[(\ell - y)_+ + (y - u)_+\right]$ penalizes misses by their size; the relative interval score is the median over images of its per-image mean over y, robust to a few extreme images while keeping the miss penalty; the share of pixels with intervals wider than $10y$ (infinite included) exposes failures a robust score hides.

Protocol. $\alpha = 0.1$ throughout. Exploratory runs (3 scene splits $\times$ 2 probe draws; E18: 20 splits; E19: 3 splits $\times$ 1 draw) chose the shrinkage $\gamma = 0.5$ (Eq. (4)), after $\gamma = 1$ showed a heavy width tail, and the design; validity holds for any fixed γ. All reported comparisons come from a confirmatory replication whose analysis plan was fixed before the runs: 20 splits $\times$ 10 draws on NYU and KITTI (plus an $r = n = 20$ arm) and in E18; in E19, 10 splits with their own anchor masks and OMNI-DC from laser anchors (Supp. G). A method is valid if its mean coverage is $\geq 1-\alpha-2\,\mathrm{SE}$ (A-dual: also within 2 SE of $r/(n+1)$), and better only if the 95% paired bootstrap interval over splits (10^4 resamples) of the IS difference excludes zero.

		Coverage		Short (%)	Δ width (%)		Δ IS (%)	
Layout	k	local	dual	dual	dual	both	dual	both
uniform	50	0.923	0.905	17.4	−49	−37	−19	−23
	200	0.901	0.905	17.4	−69	−60	−37	−39
rows	50	0.897	0.905	17.5	−41	−30	−15	−19
	200	0.873	0.905	17.5	−52	−42	−21	−26
cluster	50	0.346	0.905	17.6	−16	−1	3	0
	200	0.405	0.905	17.6	−17	−2	2	−1
edges	50	0.959	0.905	17.6	−24	−12	−4	−8
	200	0.949	0.905	17.6	−34	−24	−8	−12
near	50	0.769	0.905	17.6	−32	−18	−7	−10
	200	0.727	0.904	17.7	−37	−22	−6	−10
theory		$\geq$0.9†	0.905	17.6‡				

Table 2. **NYU and KITTI, three frozen models, $n = 20$ probes, $\alpha = 0.1$;** confirmatory, 20 scene splits $\times$ 10 probe draws; median over the six model–dataset pairs (ranges in the supplement). local: A-local calibrated on the k structured anchors themselves; dual: A-dual; both: A-dual-both. Short: % of images whose coverage on one probe draw is below 0.85. Δ: change in median relative width and relative interval score (IS, primary) against A-dist-both. †Uniform anchors only. ‡Table 1.

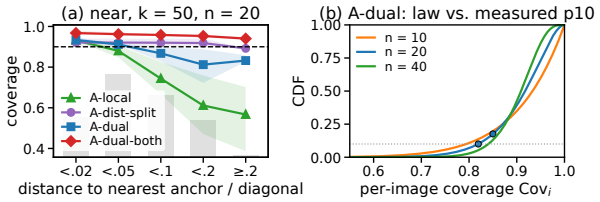


Figure 3. **What each guarantee protects.** (a) Near layout, $k = 50$, $n = 20$: A-dual's coverage decreases with distance, as Theorem 1 allows; A-dual-both is flat. Lines/bands: median/range over the six NYU/KITTI pairs; bars: pixel share; dashed: 0.9. (b) CDF of the per-image coverage law (Proposition 1), $n = 10, 20, 40$ (theory); dots, $n = 20$: measured p10 of Cov_i at 0.1 and measured share of short images at 0.85 (median over pairs, layouts, k; bars: range). 20 splits $\times$ 10 draws.

6. Results

6.1. Structured anchors break reused calibration

Calibrating within the image on the structured anchors themselves (Tab. 2, local) keeps nominal coverage only for uniform anchors. Under clustered anchors it covers 0.346 and 0.405 at $k = 50$ and 200, under near-only anchors 0.769 and 0.727, and under scan lines 0.873 at $k = 200$; edge anchors over-cover, since edges are where errors are largest. The failing intervals are narrower, not wider: with 50 clustered anchors their median relative width is 0.026–0.053 against 0.147–0.445 for uniform ones. Even uniform anchors hide a spatial gap: at $k = 50$, coverage falls from

0.93–0.94 within 0.02 of the image diagonal from an anchor to 0.73–0.88 at 0.1–0.2 of it (0.7–6.1% of pixels).

6.2. Separating the roles restores per-image validity

A-dual covers 0.904–0.905 in every layout, k and pair, at its theoretical rate $r/(n+1) = 0.905$ (within two standard errors in 56 of the 60 pair–layout–k cells), and is at least 10% narrower than A-dist-both on 6 of 6 pairs in each of the 10 layout–k cells (Tab. 2; paired bootstrap intervals exclude 0). Part of that width is A-dist-both's over-coverage (0.964–0.985), so the interval score, which charges misses, is the fairer comparison. On it A-dual is better in 46 of the 60 cells, including every cell with uniform or scan-line anchors (by 7–42%); A-dist-both is better in 13, nine of them clustered, by at most 5%. A-dual-both is better than A-dist-both in 54 cells and worse in 4, all clustered and by at most 1.1%; under clustered anchors its halves are centred differently and it gains only 0.2–4.6% in width. A-dual's coverage varies across distance bins by 3.5–12.2 points; A-dual-both reduces this to 1.1–2.3 (Fig. 3a).

6.3. The probe budget

With 20 probes the 10th percentile of Cov_i was 0.819 (law: 0.819) and 17.2–17.9% of images were short in one draw (law: 17.6%), an implementation check (Fig. 3b). The budget is fixed before any data are seen. With 10 probes A-local-split, hence A-dist-both, is infinite on every image by construction (5 calibration probes, fewer than the 9 that $\alpha = 0.1$ needs), and A-dual's law is wider (Tab. 1: p10 0.795). With 40 probes the law tightens only slowly (13.0% short, against 17.6%) at twice the measurement cost, and Sec. 4.4 shows that the nominal level is the stronger lever. We keep $n = 20$.

6.4. Which centre to calibrate

Uniform anchors. OMNI-DC is the better centre: split-CP, A-dual and A-dist-split on its output have a 16–20% lower interval score than A-dual on UniDepthV2 (paired intervals exclude zero), which fires the criterion we fixed before the run ($\geq$15%), so the layer belongs on top of the completion network. A-dual with OMNI-DC as centre costs 2.5% in interval score over split-CP on the same output and adds the per-image law: p10 0.819 and 17.8% short images in one draw, as Tab. 1 predicts. split-CP's spread is similar here (20.7% short) but carries no per-image guarantee, and it needs dense reference depth on calibration scenes. A-dist-split on OMNI-DC is the most efficient (0.071) but has no per-image guarantee and, under uniform anchors, fails with sensor-valued anchors (Sec. 6.5). Without anchors, split-CP on the frozen model has 6–7$\times$ the interval score of any anchored method.

Clustered anchors. The ranking reverses. Away from one observed patch the completion network extrapolates poorly (AbsRel 0.074, vs. 0.039 for UniDepthV2 rescaled by the same anchors; Supp. G): split-CP on its output falls short of validity (coverage 0.895, more than 2 SE below 0.9) with p10 0.676 and 23.7% short images, and A-dual on UniDepthV2 has a 41% lower interval score than A-dual on OMNI-DC (51% lower than split-CP on OMNI-DC, where A-dual beats split-CP by 17%). The monocular prior carries the far field and the anchors fix its scale; this holds over 3 splits, for one network and one dataset. On this centre A-dist-split is more efficient still (0.135), without a per-image guarantee; A-dual-both keeps both guarantees at 0.174. Across both centres and layouts, A-dual's per-image spread is the same (p10 0.818–0.820, 17.4–17.8% short), as Theorem 1 requires of any probe-independent centre.

6.5. What the probes certify: an independent sensor

Table 3b (E18, confirmatory) varies what the measurements report, for UniDepthV2 and DA-V2 (AbsRel 0.120 and 0.466) over 20 visit splits $\times$ 10 draws, on data not used to choose γ. With laser values for both, A-dual covers 0.905 with the predicted spread. With iPad anchors and laser probes, it covers the laser truth at 0.905 in both layouts with the same spread (p10 0.819, short 17.5–17.7% over both cells), because the probes measure the truth and absorb the anchors' bias into the threshold. A-dist-split, calibrated on the values the measurements report, covers it at only 0.53–0.70 under uniform anchors; its wider cluster bins absorb the bias (0.87–0.94). With laser anchors and iPad probes, A-dual over-covers (0.969–0.973 uniform, 0.920–0.926 cluster), with 2.0–2.5$\times$ wider intervals under uniform anchors, as the iPad's own errors enter the threshold. With iPad values for both, the deployed configuration without a reference, A-dual certifies the iPad reading. Under uniform anchors its intervals are about as narrow as the 1.6% sensor bias, and it covers the laser truth at only 0.566–0.632 (p10 0.11–0.18, short 68–74%); the wider cluster intervals absorb more (0.848–0.877), and the hull A-dual-both recovers the truth there (0.95–0.96) but not under uniform anchors (0.67–0.74). Calibration within the image does the same: A-local-split covers the truth at 0.909 with laser probes and 0.867–0.890 with iPad probes (Supp. H). Two methods cover the truth in every cell: split-CP, because it calibrates on laser depth of other scenes, which a deployed system lacks, and A-dist-both (at least 0.92) through width, at 1.8–2.3$\times$ A-dual's interval score under uniform anchors with laser values. Probes certify what they measure. The envelope of Supp. D restores truth coverage with iPad probes (0.993–0.996) but over-covers, at an interval score 2–78% above A-dist-both's.

7. Limitations

Expectation, not every draw: with 20 probes about 17.6% of images fall below 0.85 coverage in a single draw (Tab. 1).

(a) Laser-valued anchors and probes: which centre to calibrate

		Uniform layout					Cluster layout				
Centre	Calib.	Cov.	p10	Short %	W/y	IS	Cov.	p10	Short %	W/y	IS
UniDepthV2	split-CP	0.897	0.594	14.5	0.513	0.528	0.897	0.595	14.6	0.513	0.529
UniDepthV2 + $\mathcal{S}$	A-dist-split	0.923	0.860	7.6	0.027	0.082	0.924	0.837	12.5	0.071	0.135
	A-dual	0.904	0.818	17.7	0.030	0.089	0.905	0.819	17.6	0.109	0.182
	A-dual-both	0.949	0.905	0.8	0.038	0.087	0.967	0.930	0.5	0.126	0.174
OMNI-DC + $\mathcal{S}$	split-CP	0.897	0.813	20.7	0.024	0.073	0.895	0.676	23.7	0.336	0.374
	A-dist-split	0.923	0.850	9.9	0.026	0.071	0.924	0.757	16.7	0.391	0.422
	A-dual	0.904	0.819	17.8	0.023	0.074	0.905	0.820	17.4	0.226	0.311
	A-dual-both	0.948	0.902	1.3	0.032	0.074	0.971	0.917	2.2	0.403	0.421
theory, one draw		0.905	0.819	17.6			0.905	0.819	17.6		

(b) Laser-truth coverage by value source of anchors / probes (L = laser, I = iPad LiDAR)

		Uniform layout				Cluster layout			
Centre	Calib.	L/L	I/L	L/I	I/I	L/L	I/L	L/I	I/I
UniDepthV2 + $\mathcal{S}$	A-dist-split	0.924	0.617	0.925	0.527	0.924	0.942	0.926	0.867
	A-dual	0.905	0.905	0.973	0.566	0.905	0.905	0.926	0.848
	A-dist-both	0.979	0.968	0.978	0.920	0.974	0.978	0.970	0.958
DA-V2 + $\mathcal{S}$	A-dual	0.905	0.905	0.969	0.632	0.905	0.905	0.920	0.877
OMNI-DC + $\mathcal{S}$	split-CP	0.897	–	0.897	–	0.895	–	0.895	–
	A-dist-split	0.923	–	0.923	–	0.924	–	0.924	–
	A-dual	0.904	–	0.980	–	0.905	–	0.915	–

Table 3. **ARKitScenes,** $k = 200$ (180 **structured anchors** + 20 **probes),** $\alpha = 0.1$**.** (a) and the OMNI-DC rows of (b): E19, UniDepthV2 frozen, mean over 3 visit splits, one probe draw (confirmatory; OMNI-DC completed from iPad anchors, I/L and I/I, was not part of the confirmatory run: –). Frozen-model rows of (b): E18, mean over 20 visit splits and 10 probe draws (confirmatory). Coverage is always of the laser ground truth. p10 and short: 10th percentile of per-image coverage and % of images below 0.85, for a single probe draw. W/y: median width over laser depth; IS: relative interval score (Sec. 5). OMNI-DC sees RGB and the structured anchors only, never the probes. split-CP calibrates on dense laser depth of calibration scenes; A-dist-split on the values the measurements report. Under clustered anchors, 0.1–0.4% of OMNI-DC pixels with extreme upper bounds dominate a mean over images, not the IS shown (Supp. G). Full grid in the supplement.

No spatial guarantee from probes alone: the per-distance guarantee of A-dual-both needs calibration scenes and the same anchor rule at deployment. *What is certified:* the probes' quantity (Sec. 6.5); with sensor-grade probes our truth guarantee comes only through width: the envelope of Supp. D holds on average over scenes, not per image, and costs more than A-dist-both. ARKit depth is fused and dense; raw sparse returns and dropouts on glass or sky are untested. *The centre:* under uniform anchors a completion network is better; our centre's advantage under clusters rests on one network, one dataset and 3 splits. *Selection of* γ used NYU and KITTI, so only ARKitScenes checks efficiency on held-out data; UniDepthV2 saw ARKitScenes in training. *Scope:* one α, two budgets, one independent sensor, uniform probes by design (with a fixed-pattern scanner the guarantee covers only pixels with a return), and no downstream use: we do not show that the intervals improve an AR or planning task; active placement is not studied.

8. Conclusion

Structured sparse measurements make monocular foundation models far more accurate per image, but reused as calibration data they silently void the per-image guarantee. Giving the sensor budget two roles fixes this: structured anchors build the prediction, and a few uniformly random probes, hidden from the predictor, calibrate it within each image. The resulting layer needs no calibration images, makes the probe count a design parameter, and runs on any probe-independent centre. It certifies what the probes measure: laser probes certified the depth even with iPad anchors, iPad probes only the iPad reading.

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